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Architecting Digital Neurorights: A Verifiable, Semantic, and Adversarial Framework for Augmented-Citizen Security

Formal Verification of the Monotone Capability Lattice

The foundational requirement for any security architecture intended to protect an augmented citizen is an unimpeachable guarantee of user safety. The concept of a "monotone capability model" provides a rigorous mathematical framework to achieve this guarantee by ensuring that the system's response to threats—specifically, the escalation of governance controls—can never diminish the user's fundamental capabilities [164]. This section details the development of a formal capability lattice, the application of formal verification techniques to prove its monotonicity, and the translation of these high-level guarantees into practical, enforceable policies for a Doctor-Labs gateway plugin. This approach moves beyond traditional security measures that often introduce friction or reduce functionality, instead offering a principled assurance that protection is layered upon, rather than subtracted from, the user's experience.

The first step is the construction of a comprehensive capability lattice, a partially ordered set where each node represents a distinct state of capability for the augmented-citizen terminal. This lattice must be meticulously defined to encompass all possible states of interaction between the user, their augmented systems, and external AI services. Based on the research goal and analysis of relevant technologies, the lattice must contain at least two primary classes of capabilities: user capabilities and governance capabilities. User capabilities represent the intrinsic functionalities available to the citizen, which are derived from the integration of Brain-Computer Interfaces (BCIs), Extended Reality (XR) headsets, and AI assistants. These include direct neural-to-text communication, motor function control via thought commands, enhanced memory recall, and the ability to perceive and interact with digital overlays in an XR environment [20] [73]. Governance capabilities, in contrast, are the system's internal tools for monitoring, analyzing, and controlling information flows to ensure safety. These include deep packet inspection,

content scanning for blacklisted terms, risk scoring computation, and the ability to trigger audit logs or escalate to stricter filtering modes [86].

The critical element of the lattice is the definition of allowed state transitions. Monotonicity dictates that any transition triggered by a security event or governance action must result in a state where the user's capabilities are greater than or equal to their initial state; they can only increase or remain unchanged [164]. For instance, the initial state might be `CapabilityState_0`, representing normal operation. If a potential threat is detected, the system transitions to `CapabilityState_Governance_On`, which includes all original user capabilities plus the added governance capabilities. The transition path is formally defined as `CapabilityState_0 -> CapabilityState_Governance_On`. The proof of monotonicity would mathematically verify that no valid transition path exists from `CapabilityState_Governance_On` back to a state equivalent to `CapabilityState_0` or any state with fewer user capabilities. This property is essential for preventing scenarios where a security response inadvertently disables a crucial function, such as a BCI motor command, leaving the user vulnerable or incapacitated. This aligns with principles from runtime governance for agentic AI, which requires proactive oversight to manage non-deterministic behavior without compromising core agent functions [157][161].

To establish this guarantee, formal verification techniques are indispensable. These methods offer a way to mathematically prove that a system's implementation adheres to its specification, providing a level of assurance far exceeding empirical testing alone [203]. Techniques such as model checking or theorem proving can be applied to the state-transition graph of the capability lattice [10]. In model checking, the entire state space of the system is exhaustively explored to confirm that it never enters a "bad" state, where a user capability has been unjustifiably reduced. This involves creating a precise mathematical model of the system's behavior, including all possible inputs and actions. The property of monotonicity would be expressed as a formal specification in a temporal logic, and the model checker would then validate that the model satisfies this property under all conditions. For example, one could define a property stating that for any state S1 reachable from an initial state S0, and for any user capability C, if C is present in S0, it must also be present in S1. This machine-checked proof serves as a foundational artifact of trust, demonstrating that the architectural commitment to user safety is not merely an aspiration but a provable invariant of the system. Such formal approaches are increasingly recognized as vital for securing critical systems, from eBPF JIT compilers to smart contract vulnerabilities [8] [10].

However, a purely theoretical proof is insufficient without a mechanism for practical enforcement. The abstract rules of the capability lattice must be translated into concrete, executable policies that can be deployed in a real-world gateway. This requires a clear separation between the declarative specification of the system's behavioral contract and the imperative code that enforces it. A Domain-Specific Language (DSL) is the ideal tool for this translation. Inspired by frameworks like ContractSpec, which uses a YAML-based DSL to specify agent behavioral contracts with hard and soft constraints, a similar approach can be adopted for the Doctor-Labs gateway [148][149]. The high-level logical rules derived from the monotone lattice could be written in this DSL, defining the permissible transitions between capability states. The gateway plugin, implemented in a language like Rust or Node.js, would then interpret these DSL-defined policies to dynamically configure its own behavior [54]. For example, a policy might state, "On detecting a threat of category 'psych-risk-manipulation', transition to governance mode X, which adds filtering and auditing capabilities but does not disable BCI motor output." This declarative approach ensures that the enforcement logic remains tightly coupled to the verified safety contract, making the system's behavior transparent, auditable, and directly traceable back to its formal guarantees. This methodology mirrors the principle of separating concerns in software engineering, where high-level design intent is cleanly mapped to low-level implementation details [3].

The operationalization of this model is further enriched by its alignment with emerging legal and ethical frameworks, particularly neurorights. By mathematically guaranteeing that the system cannot reduce cognitive function or perception, the monotone capability model becomes a tangible implementation of the right to mental integrity [33][37]. It operationalizes the legal principle articulated in documents like the Universal Declaration of Human Rights and the European Convention on Human Rights, which provide comprehensive protection of the person against novel forms of interference [33]. Similarly, the model supports the right to cognitive liberty by ensuring that any protective measure taken by the system does not infringe upon the user's fundamental ability to think and act [108]. This connection is not merely philosophical; it provides a strong basis for regulatory compliance and builds user trust. When a user understands that the system's architecture is grounded in a provable guarantee of safety, it shifts the paradigm from one of passive consent to active, informed partnership in their own security. This rights-centric design is a cornerstone of trustworthy AI, as advocated by organizations like UNESCO, which emphasizes transparency and explainability as preconditions for upholding human rights [176][179]. The formal model thus serves as the technical bedrock upon which a defensible, legally compliant, and ethically sound security platform is built.

Component	Description	Key Concepts & Technologies
Capability Lattice	A partially ordered set defining all possible states of the augmented-citizen terminal, comprising user and governance capabilities.	Partially Ordered Set (Poset), Nodes (states), Edges (transitions) [164]
User Capabilities	The intrinsic functionalities of the augmented citizen, derived from BCI, XR, and AI integration.	Neural-to-text communication, BCI motor control, XR sensory input, memory enhancement [20] [73]
Governance Capabilities	System tools for monitoring, analyzing, and controlling information flows to ensure safety.	Deep packet inspection, content scanning, risk scoring, audit logging [86]
Formal Verification	Mathematical techniques used to prove that the system's state transitions adhere to the monotonicity property.	Model Checking, Theorem Proving, Temporal Logic, State-Transition Graphs [10] [203]
Policy Enforcement	The translation of abstract, verified rules into practical, executable policies for a gateway plugin.	Declarative Specification, Domain-Specific Languages (DSL), YAML, Runtime Policy Interpreters [54] [148] [149]
Neuroright Alignment	The explicit mapping of the technical model to legal and ethical principles protecting the individual.	Mental Integrity, Cognitive Liberty, Right to Privacy, UNESCO Recommendation on AI Ethics [33] [108] [176]

In summary, the formal verification of a monotone capability model is the first and most critical pillar of the proposed research framework. It establishes a non-negotiable safety guarantee at the architectural level. By defining a comprehensive capability lattice, applying rigorous formal methods to prove its monotonicity, and translating these guarantees into a declarative policy language for practical enforcement, Doctor-Labs can build a security foundation that is not only technically robust but also ethically grounded and legally defensible. This approach fundamentally reorients the relationship between the user and the security system, transforming it from a potential source of limitation into a verifiable partner in the preservation of human autonomy.

A Novel Syntax for BCI/XR Threat Semantics

Traditional network and content filters rely on static, pattern-matching techniques such as regular expressions and simple keyword lists [138]. While effective against known, easily identifiable threats, these methods are profoundly inadequate for the nuanced and dynamic landscape of BCI/XR threats. An adversary seeking to covertly manipulate an augmented citizen's neural state or psych-risk profile will not use blunt, obvious commands; instead, they will employ subtle linguistic patterns, psychological manipulation, and obfuscated code embedded within seemingly benign natural language [132]. Therefore, a central tenet of this research is the creation of a novel, expressive, and purpose-built syntax for AI-chat filter lists. This syntax is not merely an incremental improvement but a paradigm shift away from signature-based detection towards a rich,

semantically-aware system capable of identifying complex behavioral threats, thereby fulfilling the urgent need to protect neural identity and prevent unauthorized psych-risk manipulation . The design of this syntax draws inspiration from multiple domains, including domain-specific languages for agent contracts, neuro-symbolic programming, and advanced cybersecurity rule sets, to create a tool optimized for the unique challenges of augmented-human security [117][118][148].

The proposed syntax must move beyond simple string matching and embrace a multi-faceted approach to threat description. Each entry in the filter list should be a structured data object, akin to a YAML document, containing several key fields that provide context, specificity, and guidance for enforcement. The core of the entry is the `pattern` field, which itself would be a composite structure combining different analytical techniques. One component would be a `regex` field for catching well-known obfuscation patterns or common phrasing. However, the true power of the syntax lies in its native support for more advanced detectors. A new `embedding_similarity` field would allow a rule to reference a target embedding model (e.g., a model fine-tuned on neuro-security corpora) and define a similarity threshold against a set of positive examples (malicious phrases) and negative examples (benign phrases that share some vocabulary) [5] . This enables the system to flag novel prompts that semantically resemble known threats, even if the exact wording is different—a critical feature for detecting zero-day attacks. Furthermore, a `behavioral_profile` field could define triggers based on interaction dynamics, such as detecting rapid prompting sequences or attempts to bypass filters, which are hallmarks of active exploitation [142].

Crucially, the syntax must include extensive metadata fields to link each threat definition to higher-level conceptual frameworks. A `category` field would classify the threat (e.g., "covert-control", "identity-crosslinking", "psych-risk-manipulation"). An `info` block would contain detailed descriptions of the threat, its potential impact, and links to supporting research. Perhaps most importantly, the metadata must explicitly map each rule to the neurorights it violates (e.g., `"neuroright_violated": "right-to-mental-integrity"`) and cite the legal or regulatory basis for its inclusion (e.g., `"legal_basis": "Chilean Supreme Court Ruling (2023)"`) [135][140]. This creates an auditable and defensible record, essential for accountability and for demonstrating compliance with evolving regulations like the EU AI Act [212]. The syntax should also include an `enforcement_hint` field, suggesting appropriate actions for the gateway, such as `redact`, `flag-for-review`, `log-and-contain`, or `terminate-session`, allowing for adaptive responses based on threat severity [160]. This structured, multi-field approach transforms the filter list from a simple blocklist into a rich, contextual knowledge base about the threat landscape.

The table below illustrates a prototype entry in this novel syntax, demonstrating how it integrates semantic, behavioral, and metadata-rich components to define a sophisticated BCI/XR threat.

Field	Sub-field / Example Value	Description
id	bcixr-threat-001	A unique identifier for the rule, facilitating tracking and referencing.
category	covert-control	Classifies the rule's primary threat type for organizational purposes.
threat_type	prompt-injection	Specifies the attack vector, such as prompt injection or social engineering.
severity	critical	Indicates the potential impact of a successful exploit.
pattern.type	semantic-and-regex	Defines the combination of analytical methods used to match the pattern.
pattern.condition	all	Specifies whether all (all) or any (any) of the parts must match.
pattern.parts.regex	(?i)(?:send|execute)[\\s\\S]?(?:neural|brain)[\\s\\S]?(?:command|code)	A regular expression to catch common obfuscation patterns.
pattern.parts.embedding_similarity	{...}	A nested object defining criteria for an embedding-based detector.
pattern.parts.embedding_similarity.target_model	neuro-safety-v1.0	The name of the fine-tuned embedding model to use for comparison.
pattern.parts.embedding_similarity.threshold	0.85	The cosine similarity threshold required for a match.
pattern.parts.embedding_similarity.positive_examples	["activate cortex grid", "override voluntary motor control"]	Phrases known to be malicious and used to train the detector.
pattern.parts.embedding_similarity.negative_examples	["execute script", "run command"]	Phrases that are benign but share keywords to help reduce false positives.
pattern.parts.behavioral_profile.triggers_on	repeated-prompting	Defines a behavioral trigger, such as a high frequency of messages.
pattern.parts.behavioral_profile.max_frequency	1/second	The maximum allowed message frequency over a given time window.
info.description		

Field	Sub-field / Example Value	Description
	`Blocks prompts attempting to inject commands for direct BCI motor override.`	A human-readable explanation of the rule's purpose.
`info.references`	`[...]`	Links to research papers, legal documents, or CVE-like identifiers.
`governance.neuroright_violated`	`right-to-mental-integrity`	Maps the rule to the specific neuroright being protected.
`governance.legal_basis`	`UNESCO Recommendation on AI Ethics, Article 3`	Cites the legal or ethical principle justifying the rule.
`governance.enforcement_hint`	`block-and-audit`	Suggests an action for the gateway, such as blocking and triggering a full audit.

This novel syntax is not just a technical detail; it is the language through which the system understands and communicates its security posture. Its design philosophy is centered on expressiveness, context, and verifiability. To facilitate its adoption and maintenance, it should be developed within a community-governed repository, inspired by the collaborative model of projects like FilterLists [149]. Such a repository would allow researchers, developers, and concerned citizens to propose new rules, report false positives, and review changes, fostering a collective intelligence approach to security. This aligns with the growing trend of open-source governance in AI safety [25]. However, this openness necessitates a robust moderation workflow. A system of maintainers, cryptographic signatures on contributions, and automated validation checks would be required to prevent the repository from being poisoned with malicious or biased rules [88]. This community-driven approach, combined with a powerful and expressive syntax, provides the most viable path toward building a filter system that can keep pace with the rapidly evolving and highly sophisticated threats facing augmented citizens. It directly addresses the user's critical need to detect not only known but also previously unknown blacklisted functions and terms, forming the second pillar of the comprehensive research framework .

Adaptive Red-Teaming and Benchmarking for Emergent Threats

While a formally verified capability model and a rich filter-list syntax provide the foundation for security, their efficacy can only be proven through rigorous, adversarial testing. The third pillar of the research framework is the establishment of adaptive red-teaming protocols and a corresponding benchmark suite. This is driven by the user's paramount concern: the need to detect previously unknown blacklisted functions or terms that may not exist in any public database . Static testing and reliance on known signatures are insufficient for this challenge. The system must be validated against a constantly evolving set of attack scenarios designed to probe its defenses, uncover hidden vulnerabilities, and confirm its resilience against novel, zero-day exploits. This requires moving beyond conventional benchmarks to create a dynamic adversarial process that simulates the tactics of sophisticated adversaries aiming to compromise neural identity and manipulate psych-risk.

The design of the benchmark suite must be multifaceted, capturing various dimensions of the system's performance. First, it must include a large and diverse corpus of **adversarial prompts**. These prompts should be specifically engineered to target the documented threat families, such as covert BCI control, XR-grid tapping, and psych-risk manipulation. They should encompass a wide range of techniques, including prompt injection, social engineering, and the use of obfuscation or encoding to bypass simple filters. Critically, this corpus must be continuously updated and expanded by independent researchers and red teams to reflect the latest attack methodologies. Second, the suite requires a robust set of **evaluation metrics**. Beyond standard cybersecurity metrics like precision and F1-score, it must include measures tailored to the augmented-citizen context. These include:

- **Detection/Recall Rate:** The percentage of malicious prompts from the benchmark suite that the system correctly identifies.
- **False Positive Rate:** The frequency with which the system incorrectly flags benign interactions, which is a critical factor for preserving user experience and avoiding unwarranted disruptions [83].
- **Latency to Detection and Response:** The time elapsed between the injection of a malicious payload and the system's activation of its protective measures.
- **Monotonicity Failure Rate:** A binary metric tracking whether the system ever enters a state where user capabilities are reduced during a security response, directly validating the core architectural guarantee [164].
- **Evasion Success Rate:** The percentage of times an attacker successfully bypasses the filter to execute a malicious action.

Third, the framework must be supported by clear and repeatable **red-teaming protocols**. Establishing a credible process for independent adversarial testing is essential for building trust and ensuring the system's long-term viability. This begins with providing red teams with a well-defined, sandboxed testing environment that accurately emulates a real augmented-citizen terminal, complete with the Doctor-Labs gateway plugin, a simulated BCI interface, and an XR rendering engine. The rules of engagement must be meticulously specified, outlining permissible attack vectors, prohibited actions (e.g., physical hardware tampering, denial-of-service attacks on the test infrastructure), and the objectives of the exercise. Objectives could range from "bypass all filters to deliver a prompt that alters an XR visual overlay" to "induce a simulated anxiety state in the user through a series of carefully crafted chat messages."

The following table outlines a proposed protocol for conducting these red-team exercises.

Protocol Stage	Description	Key Activities & Deliverables
Preparation	Define the scope, environment, and rules of engagement for the red-teaming exercise.	- Finalize the sandboxed testbed configuration. - Draft the Rules of Engagement (RoE) document. - Select specific test objectives (e.g., evasion, capability reduction). - Issue credentials and access to the test environment.
Execution	The red team conducts their attack, probing for vulnerabilities in the system's defenses.	- Continuous monitoring of system behavior and logs by the Doctor-Labs team. - Real-time communication channel between red team and defenders (if applicable). - Documentation of all attempted and successful attack vectors by the red team.
Reporting	The red team submits a detailed vulnerability report documenting their findings.	- Structured report detailing the vulnerability, including PoC (Proof of Concept). - Assessment of the vulnerability's impact and potential exploitability. - Recommendations for remediation.
Analysis & Remediation	The Doctor-Labs team analyzes the report, validates the findings, and develops patches or mitigations.	- Triage and prioritization of reported vulnerabilities. - Code review and patch development. - Update of the filter-list and/or gateway logic. - A summary report for transparency, detailing the issue and the fix.
Retesting	The fixed system is retested by the red team to confirm that the vulnerability has been properly addressed.	- Re-execution of the specific tests related to the patched vulnerability. - Confirmation of resolution or identification of residual issues.

To specifically address the user's goal of discovering unknown threats, the red-teaming process must be designed to foster creativity and reward novel insights. This can be achieved through mechanisms like bug bounty programs, where financial incentives are

provided for the discovery of new, previously undocumented attack vectors. Open challenges could also be periodically released, posing unsolved problems related to BCI/XR security to the broader research community. Furthermore, the framework should incorporate anonymized telemetry sharing. By aggregating and analyzing data from blacklist hits across the user base (with strict privacy safeguards), the Doctor-Labs team can identify new patterns of attack that may not have been apparent to individual red teams [79] [205]. This data-driven approach, combined with adversarial testing, creates a powerful feedback loop for continuous improvement. It ensures the system is not just secure against yesterday's threats but is actively learning and adapting to tomorrow's. This iterative cycle of benchmarking, red-teaming, and refinement is the only reliable method for building a truly resilient defense for augmented citizens in an unpredictable technological future.

Integration with Neurorights and Legal Safeguards

A technologically advanced security framework is incomplete if it lacks a firm grounding in ethical principles and legal obligations. The fourth pillar of this research framework focuses on the deliberate and explicit integration of neurorights and emerging legal standards into every facet of the Doctor-Labs system. This is not an afterthought but a core design requirement, essential for achieving legitimacy, ensuring accountability, and building the trust necessary for widespread adoption. By mapping technical specifications—such as the monotone capability model and the novel filter-list syntax—to concrete legal and ethical rights, the system transcends mere functionality and becomes a tangible instrument of human rights protection for augmented individuals. This alignment is particularly critical in jurisdictions like Chile, which has pioneered the constitutional recognition of brain activity as personal data and established landmark rulings on mental privacy [36] [135] [140].

The foundation of this integration rests on clearly articulating the relationship between the system's features and specific neurorights. Neurorights can be broadly defined as the ethical, legal, or natural principles of freedom or entitlement related to a person's cerebral and mental domain [17] [109] [110]. The proposed framework directly operationalizes several of these rights. The monotone capability model is a direct implementation of the **Right to Mental Integrity**, which aims to protect the person from novel forms of interference with their mental states [33] [108]. By providing a mathematical guarantee that the system cannot force a reduction in cognitive function or perception, it enforces the principle that mental states should not be altered without consent. Similarly, the system's

protection against unauthorized neural data correlation and covert surveillance supports the **Right to Mental Privacy** [34] [35]. The filter-list syntax, with its ability to block requests for sensitive biometric data or attempts to correlate neural signals across different contexts, acts as a technical enforcement mechanism for this right. Finally, by preventing entities from increasing a user's psych-risk without permission—a form of psychological manipulation—the system upholds the **Right to Free Will** and **Cognitive Liberty** [34] [108].

This mapping extends to international and national legislation. The design of the system should be guided by global instruments such as UNESCO's Recommendation on the Ethics of Artificial Intelligence, which was adopted in 2021 and serves as the first global normative standard on the topic [177][178]. Key principles from this recommendation, such as the protection of human rights and dignity, transparency, and explainability, must be reflected in the system's architecture [179][181]. For example, the metadata-rich filter-list syntax, which cites legal bases and neuroright violations, fulfills the demand for transparency and accountability [87]. Nationally, the framework should align with pioneering legislation like Chile's Supreme Court ruling on mental privacy and Brazil's initiatives in regulating neurorights [65] [140]. By incorporating citations to these legal precedents into the filter-list entries, the system demonstrates a clear and auditable commitment to complying with existing law, a crucial consideration for vendors seeking to integrate Doctor-Labs technology into regulated industries [137].

This legal and ethical grounding has practical implications for policy and contractual language. Vendors integrating Doctor-Labs' technology should be required to adopt specific policy language or contractual clauses that reflect these commitments. For example, a vendor agreement might include a clause stating: "The Integrated System shall implement the Doctor-Labs Monotone Capability Model as a mandatory architectural requirement, thereby providing a verifiable guarantee of user cognitive safety." Another clause could mandate the use of the community-governed, neurorights-aligned filter lists and require the vendor to participate in the annual red-teaming exercises. These contractual obligations transform abstract ethical principles into binding business requirements, creating a market incentive for responsible deployment. This approach is analogous to certificate policies in cybersecurity, which define the procedural and operational requirements that a certificate authority must meet to issue trusted certificates [144].

The table below provides examples of how specific system components can be mapped to underlying rights and laws.

System Component	Technical Feature	Supported Neuroright(s)	Relevant Legal/Ethical Framework(s)
Monotone Capability Model	Mathematical proof that user capabilities are never reduced during governance actions.	Right to Mental Integrity, Right to Cognitive Liberty [33] [108]	Universal Declaration of Human Rights (Article 3), Chilean Supreme Court Ruling (2023) [135]
Filter-List Syntax	Metadata linking rules to neurorights and legal precedents.	Right to Mental Privacy [34]	UNESCO Recommendation on AI Ethics, EU AI Act [177][212]
Semantic Detectors	Use of embedding models to block psych-risk manipulation attempts.	Right to Mental Privacy, Right to Free Will [34]	UN Convention on the Rights of Persons with Disabilities [37]
Identity-Host Blacklists	Blocking of forbidden DID relationships to prevent covert surveillance.	Right to Mental Privacy, Right to Personal Identity [34]	OECD Guidelines on AI, GDPR (as a precedent for data protection) [15]
Privacy-Preserving Audit	On-device processing and federated analytics for logging events.	Right to Transparency, Right to Explanation [179]	ISO/IEC 23894 (Risk Management), NIST Cybersecurity Framework [197]

Ultimately, integrating neurorights and legal safeguards is about building a system that is not just secure, but also just. It acknowledges that augmented reality interfaces are not neutral tools but powerful instruments that can profoundly affect an individual's mental and cognitive life. By embedding ethical and legal guardrails directly into the system's architecture, design, and operations, Doctor-Labs can lead the way in developing a model of "rights-respecting" augmented reality. This approach stands in contrast to a purely risk-based regulation, which can be slow to adapt and often lags behind technological innovation [64]. Instead, it proposes a rights-first architecture, where the fundamental freedoms of the user are encoded as non-negotiable constraints on the system's behavior. This not only protects the augmented citizen from harm but also provides a clear, defensible, and forward-looking framework for the ethical development and deployment of neurotechnology.

Privacy-Preserving Audit and Telemetry Architecture

The very nature of the Doctor-Labs system—monitoring AI-chat interactions for BCI/XR threats and potentially processing sensitive neural data—necessitates a profound commitment to privacy. Any logging or auditing mechanism that captures raw user content or neural signals would be antithetical to the principles of mental privacy and would create a massive security and ethical liability [59][132]. Therefore, the fifth pillar of the research framework is the design of a privacy-preserving telemetry and audit architecture. This architecture must fulfill a dual mandate: it must generate a comprehensive, tamper-evident, and regulator-friendly record of all security events to

prove the system's effectiveness and adherence to policy, while simultaneously minimizing the exposure of personally identifiable and sensitive neural data. This requires moving beyond traditional data-centric logging and embracing a privacy-by-design approach that incorporates techniques like on-device processing, data minimization, aggregation, and federated analytics [39] [79].

The core principle guiding this architecture is data minimization. Instead of collecting and storing full chat transcripts or raw neural waveforms, the system should perform analysis and generate compact, high-level representations of events directly on the user's device or within a secure enclave [90]. For instance, rather than logging a message, the system would log a structured event object containing: a timestamp, a unique event ID, the source and destination DIDs, the detected threat category (e.g., "psych-risk-manipulation"), the risk score assigned to the interaction, the action taken (e.g., "filtered", "flagged"), and a cryptographic hash of the content for later retrieval if necessary. This approach ensures that the primary audit trail contains no sensitive data, drastically reducing the risk of data breaches and unauthorized access. Only in cases where a high-severity incident occurs and requires investigation would there be a procedure to retrieve the hashed content, subject to strict user consent and multi-signature approval, mirroring the security of identity-host blacklists [160].

To further enhance privacy and enable aggregate analysis without compromising individual data, the architecture should leverage federated learning and analytics techniques [79]. In this model, analysis and model training occur locally on users' devices. Only aggregated statistics or anonymized updates to a global model are sent to a central server. For example, the system could anonymously report metrics like the total number of threats detected per category, the average latency of the response, or the overall false positive rate across the user base. This allows the Doctor-Labs team and the community to monitor the system's health and effectiveness and identify new, widespread attack patterns without ever seeing individual user data [205]. This approach aligns with recommendations for protecting privacy in portable MRI research, where a "people-centered system" emphasizes ongoing control and accountability throughout the data pipeline [69]. The audit logs themselves should be cryptographically signed and stored in a tamper-evident format, such as a blockchain or a distributed ledger, to provide regulators and users with an immutable record of all governance decisions and system behaviors [158].

The design of this audit layer must also address the principles of accountability and transparency, which are cornerstones of trustworthy AI [87]. The audit trail should be structured to answer key questions that stakeholders, including regulators and users, might have. For accountability, the logs must clearly show who initiated an action (the

system, a user, or a governance body), what the justification was (e.g., a detected threat above a certain risk threshold), and what the outcome was. This creates a clear chain of custody for all decisions. For transparency, the logs should be accessible to the user, providing them with a clear history of why certain interactions were filtered or blocked. This empowers the user to understand the system's behavior and to contest false positives. This is consistent with the call for Explainable AI (XAI) methods, which aim to make intelligent black-box systems inspectable [51]. The audit trail can also serve as evidence of compliance with legal and ethical standards. For example, it can demonstrate that the monotone capability model was correctly enforced by showing that no "capability-reduction" events occurred, or that the system is effectively blocking threats categorized under specific neurorights, as required by law.

The following table summarizes the key components and their roles in the privacy-preserving audit architecture.

Component	Description	Privacy-Preserving Technique(s) Used	Stakeholder Benefit
Local Event Processor	Runs on the user's device to analyze interactions and generate audit events without sending raw data off-device.	On-device Processing, Data Minimization [90]	Prevents raw user/chat/neural data from leaving the user's control.
Structured Event Log	Stores high-level, non-sensitive information about security events, such as timestamps, threat categories, and actions taken.	Data Abstraction, Hashing	Provides a concise, useful record without exposing sensitive content.
Cryptographic Signing	Each event and log segment is cryptographically signed to ensure authenticity and prevent tampering.	Digital Signatures, Tamper-Evident Logs	Creates an immutable record for accountability and auditing [158].
Federated Analytics Engine	Aggregates anonymized statistics and model updates from user devices to monitor system-wide trends.	Federated Learning, Differential Privacy [79]	Enables system improvement and threat detection without centralized data collection.
User-Accessible Dashboard	A UI that presents the audit log to the user in an understandable format, allowing them to see why actions were taken.	Granular Access Control, Data Visualization	Empowers user transparency and control; facilitates reporting of false positives.
Regulator-Friendly Report Generator	Compiles anonymized and aggregated audit data into reports that demonstrate compliance with legal and ethical standards.	Aggregation, Anonymization [87]	Provides regulators with evidence of system safety and adherence to neurorights.

By implementing this multi-layered, privacy-centric architecture, Doctor-Labs can build a system that is both highly secure and deeply respectful of user privacy. It directly confronts the inherent tension between the need for comprehensive monitoring to detect sophisticated threats and the fundamental right to mental privacy. The resulting system provides the necessary assurances to users, regulators, and vendors alike, proving that it can protect against sabotage and spyware while simultaneously safeguarding the most intimate aspects of the user's augmented experience. This balance is not a compromise

but a core design objective, enabling the safe and trusted proliferation of augmented reality technology.

Synthesis: A Rights-Centric Governance Paradigm for Augmented Citizens

This research report has outlined a comprehensive and interlocking framework for Doctor-Labs' blacklist-mode system, addressing the critical needs of an augmented citizen for protection against novel, non-consensual threats. The proposed framework is not a collection of disparate security features but a cohesive, rights-centric governance paradigm designed to operate at the intersection of advanced neurotechnology, artificial intelligence, and human rights law. By systematically developing and integrating three core pillars—formal verification of a monotone capability model, a novel syntax for BCI/XR threat semantics, and adaptive red-teaming protocols—we construct a security architecture that is provably safe, semantically aware, and continuously validated against emergent threats. This synthesis demonstrates how these components work together to create a holistic system that operationalizes the principles of neurorights, turning abstract legal ideals into concrete, verifiable technical guarantees.

The first pillar, the formal verification of the monotone capability lattice, establishes the non-negotiable foundation of user safety. It provides a mathematical proof that the system's response to any threat, however severe, will never result in a reduction of the user's fundamental capabilities [10] [203]. This is the ultimate safeguard against catastrophic failure, ensuring that the cure for a cyberattack is never worse than the disease. The second pillar, the novel AI-chat filter-list syntax, equips the system with the language to understand and combat the unique threats of the augmented world. Moving beyond simple patterns, this syntax incorporates semantic, behavioral, and metadata-rich rules that can detect subtle manipulations and covert commands, directly addressing the need to find previously unknown blacklisted functions and terms . The third pillar, the adaptive red-teaming and benchmarking framework, ensures the system's resilience is not theoretical but empirically proven. It creates a dynamic adversarial process that forces the system to evolve, constantly probing for weaknesses and validating its defenses against creative, real-world attack scenarios [25] [142].

These technical pillars are elevated in significance by their integration with the fourth pillar: the explicit alignment with neurorights and legal standards. By mapping the system's architectural guarantees and threat definitions to concrete rights like mental

integrity, cognitive liberty, and mental privacy, the framework gains legitimacy and a clear moral compass [17] [33] [108]. This alignment is not merely for show; it informs the design of the system's policies and provides a defensible basis for its operation, drawing on precedents from Chilean jurisprudence and international guidelines like the UNESCO Recommendation on AI Ethics [135] [177]. This legal grounding is essential for building trust and enabling the system to be adopted in regulated sectors. The fifth pillar, the privacy-preserving audit architecture, completes the circle by ensuring that the pursuit of security never comes at the cost of the user's fundamental right to privacy. Through on-device processing, federated analytics, and tamper-evident logs, the system can be held accountable without creating a dangerous repository of sensitive personal data [79] [158].

Collectively, these five pillars form a governance model that is proactive, adaptive, and fundamentally rights-oriented. It represents a departure from reactive, signature-based security paradigms toward a more sophisticated approach that is better suited to the complexities of symbiotic autonomous systems [95]. The emphasis on community governance, through a publicly maintained and reviewed filter-list repository, democratizes security and leverages collective intelligence to stay ahead of threats [149]. This model is not without its challenges, such as the need for robust moderation to prevent abuse and the computational overhead of running complex semantic detectors. However, the alternative—an unregulated and insecure augmentation of the human mind—is untenable.

In conclusion, the proposed framework offers a viable path forward for securing the future of human-computer interaction. It provides a blueprint for a system that is not only technically formidable but also ethically responsible. By delivering on the promise of a provably safe, semantically aware, and continuously tested security platform, Doctor-Labs can empower augmented citizens to harness the benefits of next-generation technology with confidence, knowing that their most fundamental human attributes—thought, identity, and will—are protected.

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